

# The Carry-Over Lottery Allocation Family: Bounds, Backtests, and a Configuration Space for NBA Draft Reform

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## Abstract

The annual NBA draft assigns 30 indivisible picks to 30 teams. The top picks go through a lottery for the 14 non-playoff teams, and the lottery has a documented manipulation incentive (“tanking”) a team eliminated from playoff contention can improve its expected draft position by losing additional games. Banchio and Munro [1] prove that no mechanism based on end-of-season records alone can simultaneously reward poor performance and eliminate this incentive. Carry-Over Lottery Allocation (COLA) sidesteps the impossibility by replacing single-season records with multi-year playoff history as the basis for draft priority [2]. We formalize the COLA family as a configuration space parameterized by seven design dials (eligibility, increment, diminishment, lottery weighting, cap, carry-over scope, tiebreak). Each variant proposed in Highley et al. [2] and the Substack series is a point in this space, and the NBA’s 2026 “3-2-1” proposal sits at a degenerate corner. We instantiate the framework against the ZenGM Basketball-GM engine, grid-searching 48 configurations (eligibility  $\times$  cap  $\times$  carry-over scope) under 50 replicates of 30 simulated seasons. Under universal-dominance comparison on three league-relevant objectives (maximum years between conference-finals appearances, manipulation gain, rank-1-to-5 lottery spread), Capped COLA at MAX = 150 is the only named variant on the Pareto frontier; Status quo, Classic, Simple, and 3-2-1 are all dominated. Three of the five Pareto-optimal configurations adopt reset-on-championship carry-over scope. As supporting analytical results we derive a closed-form  $\sim 4\%$  upper bound on the worst-case probability gain from pool manipulation in the Classic configuration (Theorem 2) and a per-series tanking bound for the Capped configuration (Lemma 3). A corollary shows that the cap value  $C$  drops out of the Capped manipulation-gain bound, which depends only on  $|E|$ :  $\Delta p_{\text{capped}} \leq 0.3/|E|$  in the play-in worst case. The empirical sweep confirms this cancellation across  $C \in \{100, 150, 200\}$ . A 26-season historical backtest (1999–2025) reproduces the Philadelphia “Process” as a second-tier validation alongside the headline Pareto result.

## 1 Introduction

In the 2025-26 NBA season, by some counts roughly a third of the league’s 30 teams were openly losing games at season’s end to improve their lottery odds. The league responded with several reform ideas, including the “3-2-1 lottery” [3]. The proposal carries a three-year re-evaluation provision: the design discussion will reopen in 2029.

The annual NBA draft assigns thirty indivisible draft picks to thirty teams. Picks 15–30 go by reverse regular-season record. Picks 1–14 go through a lottery confined to the 14 non-playoff teams, with weighted odds tilted toward teams with worse records (in the current system, the bottom three teams are weighted

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equally and the rest decline). The NBA has adjusted those odds repeatedly while facing the same question: how to assign indivisible picks while (a) preferentially rewarding the weakest teams, (b) eliminating the incentive to under-perform, and (c) keeping the mechanism transparent enough that fans, owners, and players can reason about it.

Banchio and Munro [1] give a formal treatment of the central tension and prove an impossibility. They define a *No-Tanking Condition* (NTC): no team can improve its expected outcome by losing. Any mechanism mapping end-of-season records to draft probabilities while giving worse-record teams strictly better odds must violate NTC. Only a uniform lottery satisfies NTC, abandoning differential reward. The 2026 “3-2-1” proposal takes precisely this route, with adjustments for the play-in tournament and reduced odds for the three teams at the bottom [3].

Carry-Over Lottery Allocation (COLA) sidesteps the impossibility on a different axis: it replaces single-season records with multi-year playoff history as the basis for draft priority [2]. Each non-playoff team accumulates draft priority over consecutive seasons; success in the playoffs or the draft itself diminishes that priority. The single-season manipulation channel that drives the Munro-Banchio counterexample becomes ineffective: no single-season choice materially shifts the multi-year ticket pool. COLA is a family rather than a single design; the original paper develops Classic COLA, and subsequent work introduces variants at different points in the design space.

## Contributions:

- **(§3) Seven-dial configuration space.** We formalize the COLA family as a 7-tuple  $(E, \Delta, \rho, W, C, S, T)$ : eligibility, increment, diminishment, lottery weighting, cap, carry-over scope, tiebreak (Definition 2). Each named variant (Simple, Simple Lottery, Classic, Countdown, Capped) is a point in this space; the NBA’s 2026 “3-2-1” proposal sits at a degenerate corner. Each dial maps to a downstream league-relevant property, converting variant comparison from prose to tuple comparison.
- **(§5) Basketball-GM Pareto frontier.** We compute the Pareto frontier in a 48-configuration grid (eligibility  $\times$  cap  $\times$  carry-over scope) under 50 replicates of 30 simulated seasons, on three objectives: maximum years between conference-finals appearances (lower better), manipulation gain (lower), and rank-1-to-5 lottery spread (higher). Capped COLA at  $\text{MAX} = 150$  is the only named variant on the frontier; Status quo, Classic, Simple, and 3-2-1 are dominated.
- **(§4, §5) Analytical bounds and empirical confirmation.** A closed-form  $\sim 4\%$  upper bound on pool-manipulation gain under reasonable index distributions (Theorem 2), plus a per-series tanking bound for Capped (Lemma 3). Corollary 4: the cap value  $C$  drops out of the Capped manipulation bound; only  $|E|$  matters. The sweep confirms the cancellation: all capped configurations realize  $\Delta p = 1.364\%$  at  $|E| = 22$ ,  $2.143\%$  at  $|E| = 14$ ,  $1.875\%$  at 16-tiered, invariant across  $C \in \{100, 150, 200\}$ . The 1999–2025 historical backtest with the 2013–2017 Philadelphia case study is a second-tier validation alongside the headline.

## 2 Background and Related Work

### 2.1 Effort Manipulation: The Munro-Banchio Impossibility

Banchio and Munro [1] prove that any mechanism mapping end-of-season records to draft probabilities, while giving the worst team strictly better odds, must violate NTC. The proof constructs a minimal counterexample: two eliminated teams with identical records and a final game between them. The loser finishes with a worse record and therefore receives better draft odds; both teams prefer to lose. Only the uniform lottery satisfies NTC, but uniform odds abandon differential reward of the weakest teams.

The authors propose *T-IC* (Truncated Incentive Compatibility) as an alternative: track each team’s conditional probability of finishing last and freeze allocations at the moment IC would first be violated. The mechanism is optimal in theory but has three practical issues. It is opaque to fans; the freeze point depends on team-level playoff-versus-draft valuations the league cannot observe; and the timing assumes teams begin the season with an incentive to win, when some teams tank from the first game.

## 2.2 Preference Manipulation and Axiomatic Limits

A second impossibility addresses preference misreporting (a team claiming to want a player it does not, to manipulate allocation order): Coreno and Balbuzanov [4] prove that no draft rule simultaneously satisfies Respect for Priority [5], Envy-Free up to One Object [6, 7], and Strategy-Proofness. The result is single-period. Whether the axioms lift to the multi-year priority structure of COLA is open and outside this paper’s scope; we note the parallel impossibility as a complement to the Munro-Banchio effort-manipulation channel that motivates the bound in Section 4.

## 2.3 Prior Reform Proposals

Three reform proposals form the immediate context. The *uniform lottery* (equal odds for all non-playoff teams) is the only mechanism satisfying NTC; the league has not adopted it. The *wheel* (a pre-committed rotation of draft positions, attributed informally to Mike Zarren) eliminates tanking by decoupling record from odds, at the cost of removing the rebuilding mechanism for weak teams. The 2026 *3-2-1 lottery* [8] expands the lottery to 16 teams across four tiers (2:3:2:1 balls per tier from the worst-record tier down) and includes the play-in losers. Structurally it is a uniform lottery with play-in adjustments and a reduction for the bottom three teams, accepting the Munro-Banchio tradeoff. COLA takes a different route: instead of flattening single-season odds, it shifts the priority basis to multi-year playoff history, an axis neither Munro-Banchio nor 3-2-1 considers.

# 3 The COLA Mechanism Family: A Configuration Space

COLA is a family of mechanisms parameterized by seven design dials. Each NBA lottery reform proposal under recent discussion (the status quo weighted lottery, the 2026 “3-2-1” proposal, and each COLA variant in Highley et al. [2] and the Substack series) corresponds to a specific choice of values for those seven dials. A policymaker who wants to move a downstream property (anti-tanking strength, weak-team rebuild horizon, broadcast complexity, multi-year predictability) is moving one or more dials. We develop the lottery index as the family’s unified state variable (§3.1), the seven dials (§3.2), the variant map (§3.3), and the dial-to-property map (§3.4). Incentive compatibility rests on five assumptions [2], summarized in §3.5.

## 3.1 Lottery Index

Every COLA variant maintains a per-team integer state across seasons and uses it to determine draft priority (DP) in the current season. We call this state the *lottery index* and write  $L_i^{(t)}$  for team  $i$  at the end of season  $t$ . Depending on the variant,  $L_i^{(t)}$  tracks drought length (consecutive seasons without a playoff series win or top-3 pick) or a stockpile of lottery tickets; both are manifestations of the same index.

**Definition 1** (Lottery Index). *For each team  $i$  and season  $t$ , the lottery index  $L_i^{(t)} \in \mathbb{Z}_{\geq 0}$  is determined by an initial value  $L_i^{(0)}$ , a per-season increment  $\Delta_i^{(t)}$  (variant-specific, depending on team performance and*

eligibility), and a per-season diminishment  $\rho_i^{(t)} \in [0, 1]$  that scales the index down based on playoff and draft outcomes:

$$L_i^{(t+1)} = \left( L_i^{(t)} + \Delta_i^{(t)} \right) \cdot \left( 1 - \rho_i^{(t)} \right). \quad (1)$$

The pool in season  $t$  is  $P^{(t)} = \sum_{i \in E^{(t)}} L_i^{(t)}$ , where  $E^{(t)} \subseteq \{1, \dots, 30\}$  is the set of teams eligible for the lottery in that season.

Three dials live in this update: the eligibility set  $E^{(t)}$ , the increment  $\Delta_i^{(t)}$ , and the diminishment  $\rho_i^{(t)}$ . A non-playoff team has positive increment (it accumulates priority); a team that wins a playoff series or a top pick has positive diminishment (it consumes some or all of that priority). §3.2 formalizes these three together with four additional dials that complete the parameterization, and §3.3 populates the 7-tuple for each variant.

### 3.2 Seven Dials

Four additional dials, implicit in the prose definitions of Highley et al. [2] and the Substack series but not yet formalized as design degrees of freedom, complete the configuration: how draft positions are assigned given the index vector, whether the index is bounded above, how long it persists across seasons, and how ties are resolved.

**Definition 2** (COLA Configuration). A COLA configuration is a 7-tuple  $(E, \Delta, \rho, W, C, S, T)$  specifying:

1. **Eligibility**  $E^{(t)} \subseteq \{1, \dots, 30\}$ : a season- $t$  predicate selecting the teams that participate in the lottery (per Definition 1).
2. **Increment**  $\Delta_i^{(t)} \in \mathbb{Z}_{\geq 0}$ : a non-negative integer added to the index of each eligible team, determined by season- $t$  state (per Definition 1).
3. **Diminishment**  $\rho_i^{(t)} \in [0, 1]$ : the fraction of the index removed after team  $i$ 's playoff or draft outcome (per Definition 1).
4. **Lottery weighting**  $W$ : a rule mapping the index vector  $L^{(t)}|_{E^{(t)}}$  to a distribution over assignments of teams to draft positions  $\{1, 2, \dots, |E^{(t)}|\}$ . The rule may be deterministic (e.g., assign by index rank) or random (e.g., draw pick 1 with probability  $L_i/P$ , then draw pick 2 from the remainder).
5. **Cap**  $C \in \mathbb{Z}_{>0} \cup \{\infty\}$ : an upper bound on  $L_i^{(t)}$ , enforced as  $L_i^{(t)} \leftarrow \min(L_i^{(t)}, C)$  after each increment.  $C = \infty$  denotes no explicit cap.
6. **Carry-over scope**  $S$ : the temporal persistence rule, one of single-season (recompute  $L_i^{(t)}$  from season- $t$  state alone, no cross-season memory), unbounded multi-year (accumulate without limit), bounded multi-year (accumulate up to  $C$ ), or reset-on-event (clear the index when a specified event occurs).
7. **Tiebreak**  $T$ : a total ordering on  $E^{(t)}$  that resolves cases where  $L_i^{(t)} = L_j^{(t)}$  for two or more eligible teams.

A COLA variant is a specification of all seven dials.

Dials 4–7 are not novel mechanisms: each variant in Highley et al. [2] and the Substack series already specifies a choice for each. Promoting them to typed dials enables cross-variant comparison via a single configuration object and dial-by-dial analysis of downstream properties. §3.3 populates the configuration per variant; §3.4 maps dials to league-relevant properties.

### 3.3 Variant Map

Table 1 populates the seven dials for the six variants under consideration: five COLA variants and the NBA’s 2026 “3-2-1” proposal, included as a degenerate configuration (§3.6). Cell values follow the verbatim specifications in Highley et al. [2], the Substack series [9, 10, 3], and Highley’s 3-2-1 reaction [8].

| Dial     | Simple                       | Simple Lottery                           | Classic                              | Countdown                               | Capped                             | 3-2-1                  |
|----------|------------------------------|------------------------------------------|--------------------------------------|-----------------------------------------|------------------------------------|------------------------|
| $E$      | 22 (non-playoff + R1 losers) | 22 (same)                                | 14 (non-playoff)                     | 22 (same)                               | exclusion pool                     | tiered 16              |
| $\Delta$ | drought +1                   | drought +1                               | $\alpha = 1,000$                     | McCarty $d \cdot w$                     | wins $w$ (play-in & below)         | none                   |
| $\rho$   | reset on series win          | reset on series win                      | 5-step, two channels                 | reset on advancement                    | 6-step + 5-step, two channels      | none                   |
| $W$      | deterministic by drought     | pre-2019 odds top-14, deterministic rest | $L_i/P$ for picks 1–4, rank for 5–14 | nested-pool MC, tickets (6, 5, 4, 3, 2) | $L_i/P$ for picks 1–5, rank for 6+ | tiered balls (2/3/2/1) |
| $C$      | $\infty$                     | $\infty$                                 | $\infty$                             | $\infty$                                | MAX = 150                          | n/a                    |
| $S$      | reset-on-event               | reset-on-event                           | unbounded multi-year                 | reset-on-event                          | bounded multi-year                 | single-season          |
| $T$      | most wins                    | subsumed by $W$                          | subsumed by $W$                      | subsumed by $W$                         | most wins (at cap)                 | subsumed by $W$        |

Table 1: The seven dials, evaluated for each variant. Dial labels:  $E$  eligibility,  $\Delta$  increment,  $\rho$  diminishment,  $W$  lottery weighting,  $C$  cap,  $S$  carry-over scope,  $T$  tiebreak (per Definition 2). The 3-2-1 proposal sits at a degenerate corner:  $\Delta$ ,  $\rho$ ,  $C$  are vacuous and  $S$  is single-season, leaving only  $E$  and  $W$  as active dials.

The variants trade off explainability, anti-tanking strength, and per-series tanking exposure differently. Simple COLA [10] replaces the lottery with a deterministic ordering on drought, extending eligibility to first-round losers; Simple Lottery COLA layers the pre-2019 NBA weighted odds on top of Simple’s drought ordering. Classic COLA is the original variant of Highley et al. [2] and the one our manipulation bound (§4) analyzes. Countdown COLA [10] uses the McCarty number (drought  $\times$  wins) and a nested-pool Monte Carlo draw that guarantees no team falls more than four spots below its priority rank. Capped COLA [3], introduced after the NBA’s reported reluctance to adopt a flexible lottery line, adds a wins-based increment, a stockpile cap (MAX = 150), and a six-step playoff diminishment schedule; the cap bounds the per-series tanking incentive (Lemma 3). The 2026 “3-2-1” proposal [8] sits at a degenerate corner: dials  $\Delta$ ,  $\rho$ ,  $C$  are vacuous,  $S$  collapses to single-season, and only  $E$  (a tiered 16-team pool) and  $W$  (a tiered ball allocation) carry the proposal’s design (§3.6).

### 3.4 Dial-to-Property Mapping

The configuration of Definition 2 reduces variant comparison to tuple comparison. The comparison is informative only when each dial corresponds to a downstream property. Four properties dominate the adoption discussion: anti-tanking strength (within-season manipulation foreclosed), weak-team rebuild help (how reliably the worst teams accumulate priority), complexity (legibility for fans and media), and predictability (multi-year forecastability for GMs and owners). Table 2 summarizes the mapping.

Three dials ( $E$ ,  $\Delta$ ,  $C$ ) bound the manipulator’s pool share  $p_{\max}$  and the size of any pool perturbation: a larger eligible set yields a larger total pool  $P$ , the increment  $\Delta$  bounds per-team per-season accumulation, and the cap  $C$  bounds the index outright. Two dials ( $\rho$ ,  $S$ ) govern multi-year accumulation: diminishment determines how quickly an index decays after success; carry-over scope determines whether accumulation

| Dial     | Anti-tanking strength                                                | Weak-team rebuild help                                                   | Complexity                                  | Predictability                                    |
|----------|----------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------|
| $E$      | Sets the pool $P$ in Theorem 2; narrower pools admit larger gain     | Determines which weak teams accumulate priority                          | High if simple count, low if exclusion-rule | Stable if record-based; less so if exclusion-rule |
| $\Delta$ | Bounds the per-season marginal-decision gain                         | Flat $\alpha$ rewards uniformly; $w$ -based rewards within-season effort | High if flat, lower if formulaic            | High if record-based                              |
| $\rho$   | Cliff size at each round transition; controls repeat-tanking returns | Restores priority to teams missing the playoffs again after success      | Medium                                      | Medium                                            |
| $W$      | Spread across rank positions; decouples position from index in part  | Tilts allocation toward worst-record teams                               | High if deterministic, low if Monte Carlo   | High if deterministic, lower if random            |
| $C$      | Bounds $p_{\max}$ and the per-series cost (Lemma 3)                  | Low cap compresses the priority signal                                   | Medium                                      | High once known                                   |
| $S$      | Escapes the Banchio and Munro [1] impossibility via multi-year scope | Multi-year scope rewards persistent weakness                             | Medium                                      | Lower under multi-year scopes (legal/contracting) |
| $T$      | Edge case for $W$ -deterministic variants                            | n/a                                                                      | High if simple                              | High                                              |

Table 2: Dial-to-property mapping. Cell entries describe the form the dial’s influence takes; the dials and the variants of Table 1 together specify how to move from one league-relevant property to another by altering one or more dial values.

persists at all (the dial whose non-single-season value escapes the Banchio and Munro [1] impossibility). The remaining two ( $W$ ,  $T$ ) translate index dynamics into draft positions. A league that finds Classic too unbounded can move  $C$  to a finite value (the Capped move). A league that finds multi-year scopes administratively heavy can move  $S$  to single-season (the 3-2-1 move). The choice among variants becomes a choice among dial settings, not among monoliths.

### 3.5 Incentive Compatibility

Highley et al. [2] establish incentive compatibility (IC) for Classic COLA under five assumptions:

1. **Playoff primacy:** a team’s value of advancing in the playoffs exceeds any plausible draft-pick gain.
2. **Playoff advancement:** within the playoffs, advancing further is preferred to losing earlier, controlling for opponent strength.
3. **Picks 5–14 negligible at the margin:** no team profits from manipulating its position among those slots through within-season effort.
4. **Pool manipulation negligible:** no team can meaningfully shift its expected draft position by altering which other teams are in or out of the pool.

5. **No traded-pick protections:** pick trades do not interact with diminishment to create manipulation incentives.

Assumptions (1)–(3) and (5) hold by construction or by short argument. Assumption (4) is supported in Highley et al. [2] by a 1,000-season simulation observing a maximum gain of roughly 3%. §4 converts that observation into a closed-form bound for the Classic configuration.

### 3.6 The 3-2-1 Proposal as a Degenerate Configuration

The NBA’s 2026 “3-2-1” proposal, slated for an owners’ vote at the end of May 2026, expands the lottery to sixteen teams across four tiers with ball counts (2, 3, 2, 1) per tier. In the configuration of Definition 2:  $\Delta$ ,  $\rho$ ,  $C$  are vacuous;  $S$  is single-season;  $E$  is the tiered 16-team pool;  $W$  is the tiered ball allocation;  $T$  is subsumed by the random draw. In Highley [8]’s framing, the mechanism is “essentially the uniform lottery but with an adjustment for the play-in and a punishment for the bottom 3 teams.” Because the worst three teams get fewer balls than the middle seven, the proposal does not satisfy the premise of the Munro-Banchio counterexample (worse teams have strictly better odds), so it sidesteps the impossibility by abandoning the rationale for differential odds. COLA takes the opposite route: it shifts the priority basis to multi-year playoff history through dial  $S$ .

## 4 Pool Manipulation Bound

### 4.1 Setup

Let  $P = \sum_{i=1}^n L_i$  denote the total lottery pool, where  $L_i$  is team  $i$ ’s lottery index and  $n$  is the number of lottery-eligible teams. A team’s probability of winning the first pick is  $p_i = L_i/P$ .

The *pool manipulation* channel operates as follows. Team  $i$  deliberately loses to a high-index opponent  $h$  near the playoff boundary, causing  $h$  to make the playoffs. Team  $h$  (with index  $L_h$ ) exits the lottery pool and is replaced by a lower-index team  $\ell$  (with index  $L_\ell < L_h$ ). The pool shrinks by  $\Delta = L_h - L_\ell > 0$ .

**Proposition 1** (Manipulation Gain). *The probability gain for team  $i$  from a pool swap of size  $\Delta = L_h - L_\ell$  is  $G_i = L_i \cdot \Delta / [P \cdot (P - \Delta)]$ .*

*Proof.* Before manipulation,  $p_i = L_i/P$ . After the swap, the new pool is  $P' = P - \Delta$  and  $p'_i = L_i/P'$ . The gain is  $G_i = p'_i - p_i = L_i(1/(P - \Delta) - 1/P) = L_i \cdot \Delta / [P(P - \Delta)]$ .  $\square$

When  $\Delta \ll P$  (typically  $\Delta/P \approx 4\%$  in our backtest), the gain factors into the manipulator’s pool share ( $p_i$ ) and the fractional pool change ( $\Delta/P$ ). Both have structural ceilings, which we exploit below.

**Indexing convention.** The bound uses  $L_i = T_i \cdot \alpha$ , where  $T_i$  is team  $i$ ’s consecutive non-playoff seasons. For Classic COLA, this is exact for any team starting from a stockpile reset (championship or top-5 lottery pick) and is the steady-state attractor under the diminishment schedule. A general bound handling ticket inheritance under the full diminishment dynamics is future work.

### 4.2 Conditioned Upper Bound

**Definition 3** (Structural Anti-Correlation). *Let  $T_{\max}$  denote the maximum consecutive non-playoff years for any team, and  $T_{\text{boundary}}$  the maximum consecutive non-playoff years for a team near the playoff boundary (a team that could plausibly be pushed into the playoffs by a single loss). The index distribution exhibits structural anti-correlation when  $T_{\text{boundary}} \ll T_{\max}$ : teams with the highest indices are far from the playoff boundary and thus cannot be the target of pool manipulation.*

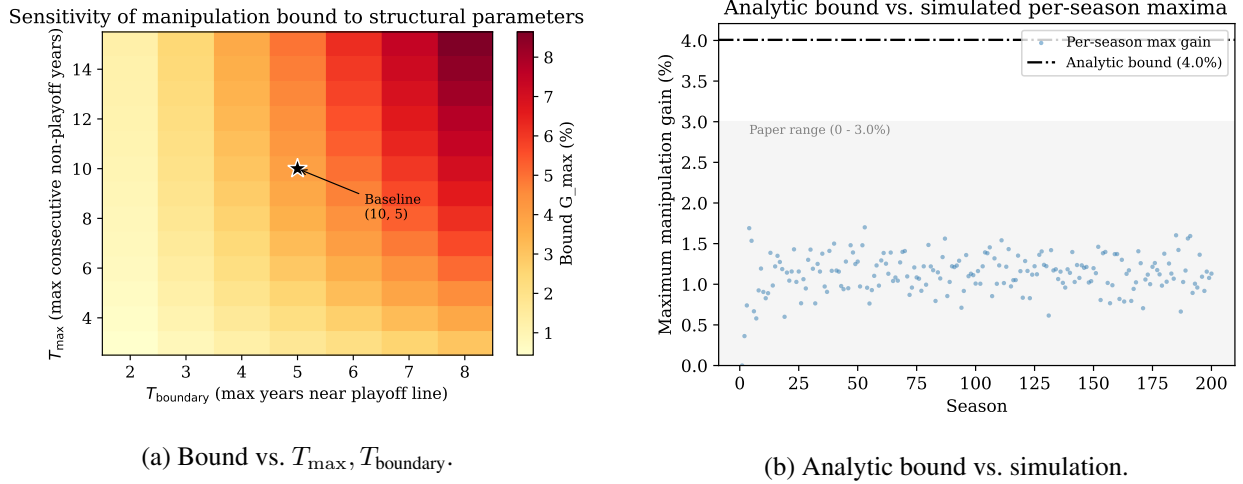


Figure 1: Manipulation bound. (a) Sensitivity to the structural-anti-correlation parameters; the bound is most sensitive to  $T_{\text{boundary}}$ , and stays under 5% for  $T_{\text{boundary}} \leq 5$ . (b) Analytic bound (red) vs. the empirical distribution of manipulation gains across 200 Bradley-Terry-simulated seasons; the bound contains all simulated outcomes with margin.

**Theorem 2** (Conditioned Manipulation Bound). *Under structural anti-correlation with parameters  $T_{\text{max}}$ ,  $T_{\text{boundary}}$ , and average index multiplier  $\bar{k} = \bar{L}/\alpha$ , where  $\alpha$  is the annual index increment, the maximum manipulation gain satisfies*

$$G_{\text{max}} \leq \frac{T_{\text{max}} \cdot (T_{\text{boundary}} - 1)}{(T_{\text{max}} + n - 1) \cdot n \cdot \bar{k}}.$$

*Proof sketch.* The gain is maximized when  $L_i$  is maximized and  $\Delta/P$  is maximized. *Bounding  $p_{\text{max}}$ :* the highest-index team has  $L_{\text{max}} = T_{\text{max}} \cdot \alpha$ . The minimum pool size occurs when one team has  $T_{\text{max}}$  years of accumulation and the remaining  $n - 1$  teams each have the minimum  $\alpha$ :  $P_{\text{min}} = (T_{\text{max}} + n - 1) \cdot \alpha$ . Thus  $p_{\text{max}} \leq T_{\text{max}}/(T_{\text{max}} + n - 1)$ . *Bounding  $\Delta/P$ :* the team being pushed out has at most  $T_{\text{boundary}} \cdot \alpha$  tickets, and the team replacing it has at least  $\alpha$ , so  $\Delta_{\text{max}} = (T_{\text{boundary}} - 1) \cdot \alpha$ . In steady state,  $P \approx n \cdot \bar{k} \cdot \alpha$ . Combining via the first-order approximation yields the stated bound.  $\square$

**Empirical calibration:** We instantiate with  $T_{\text{max}} = 10$  (typical-case ceiling on consecutive non-playoff years across teams in our 26-season backtest),  $T_{\text{boundary}} = 5$  (conservative ceiling for a near-boundary team),  $n = 14$ , and  $\bar{k} = 3.1$  (the original paper’s observed carry-over). These yield

$$G_{\text{max}} \leq \frac{10 \cdot 4}{23 \cdot 14 \cdot 3.1} \approx 4.0\%,$$

which contains the simulation-observed maximum of 3.0% across 1,000 seasons [2]. An independent 200-season Bradley-Terry simulation we ran confirms the bound (max gain 1.70%). The 26-season dataset contains one outlier (the Sacramento Kings’ 16-year drought, 2006–07 through 2021–22). Setting  $T_{\text{max}} = 16$  loosens the bound to roughly 5.1%, still above the empirical extremes; we report the typical-case bound ( $T_{\text{max}} = 10$ ) as the headline because the next-longest droughts are Minnesota at 13 years and Phoenix/Charlotte at 10.

### 4.3 Sensitivity, Coalitions, and the Pick-Value Gap

Figure 1a shows  $T_{\text{boundary}}$  as the dominant lever: under pessimistic assumptions ( $T_{\text{boundary}} = 7$ ), the bound stays under 7%; under realistic conditions ( $T_{\text{boundary}} \leq 5$ ), under 5%.  $T_{\text{boundary}}$  is also the empirical quantity most amenable to direct measurement.

Coalitional manipulation does not amplify individual gains. Joint gain for a coalition  $\{A, B\}$  is  $(L_A + L_B) \cdot \Delta / [P(P - \Delta)]$ . At 30% joint pool share, joint gain remains  $\leq 1.3\%$  and per-member gain cannot exceed the highest-index individual's. Coordination costs (detection, trust, sequential uncertainty) further dampen the channel.

The bound addresses the *probability* term of the manipulation incentive. The *value* term (the pick-value gap  $d_k - d_{k+1}$ ) has no formal ceiling in the original specification, though it is bounded for any realistic draft class. Fully closing assumption (4) would also require a bound on  $d_k - d_{k+1}$  via the distribution of draft-eligible prospect quality; we leave this to future work.

#### 4.4 Per-Series Bound for Capped COLA

Capped COLA adds a fourth manipulation channel: tanking through the playoffs (a team within the cap declining to advance, to keep its stockpile). The cap converts the playoff-diminishment schedule from a fraction-of-current-stockpile cost (unbounded as stockpile grows) to a fraction-of-MAX cost (bounded by construction).

**Lemma 3** (Per-Series Tanking Cost). *Let MAX be the Capped COLA stockpile cap. For any team entering the playoffs with  $L_i \leq \text{MAX}$ , the maximum reduction in  $L_i$  attributable to winning (rather than losing) any single playoff series is bounded by:*

$$\Delta L_{\text{series}}^{\text{max}} = \begin{cases} 0.3 \cdot \text{MAX} & \text{play-in advancers in round 1,} \\ 0.2 \cdot \text{MAX} & \text{any other series advancement.} \end{cases}$$

At  $\text{MAX} = 150$ , the bound is 45 tickets for the play-in case and 30 tickets for any subsequent series.

*Proof.* Reading off Capped COLA's six-step diminishment schedule, the per-series increments in the diminishment fraction  $\Delta\rho$  are: play-in advancer round 1 ( $\rho_{\text{lose}} = 0.1$ ,  $\rho_{\text{advance}} = 0.4$ ,  $\Delta\rho = 0.3$ ); standard round 1 ( $0.2 \rightarrow 0.4$ ,  $\Delta\rho = 0.2$ ); round 2 to conference finals ( $0.4 \rightarrow 0.6$ ); conference finals to NBA finals ( $0.6 \rightarrow 0.8$ ); finals to championship ( $0.8 \rightarrow 1.0$ ). All non-play-in transitions yield  $\Delta\rho = 0.2$ . The cost in tickets is  $\Delta\rho \cdot L_i^{\text{pre}} \leq \Delta\rho \cdot \text{MAX}$ , with equality at the cap.  $\square$

The bound makes the playoff-tanking incentive transparent: at  $\text{MAX} = 150$ , no single series win ever costs more than 45 tickets, roughly a year and a half of accumulation for a team near the wins-increment ceiling. The cumulative regret for a finals-losing team at the cap is  $0.8 \cdot \text{MAX} = 120$  tickets, decomposed per Lemma 3.

**Corollary 4** (Cap-Cancels-Out for Capped Configurations). *For a Capped COLA configuration with eligibility pool  $E$  of size  $|E|$ , cap value  $C$ , and per-series-cost upper bound  $\kappa \cdot C$  (where  $\kappa = 0.3$  in the play-in worst case of Lemma 3 and  $\kappa = 0.2$  for any subsequent series), the worst-case manipulation-gain bound for a single-series pool perturbation is*

$$\Delta p_{\text{capped}} \leq \frac{\kappa \cdot C}{C \cdot |E|} = \frac{\kappa}{|E|}.$$

The cap value  $C$  drops out. The manipulation-gain bound depends only on  $\kappa$  and  $|E|$ . The sweep in §5.3 confirms the cancellation empirically.

*Proof.* The maximum index a single team can hold under Capped COLA is  $C$ , and the worst-case pool of  $|E|$  such teams is  $C \cdot |E|$ . The maximum single-series perturbation in a team's index from the pool is  $\kappa \cdot C$  tickets, by Lemma 3. Substituting into the gain expression  $\Delta = (\text{perturbation})/(\text{pool})$  yields  $\kappa \cdot C / (C \cdot |E|) = \kappa / |E|$ .  $\square$

The cap controls per-team accumulation speed and per-series tanking cost (Lemma 3); it does not control pool-wide manipulation resistance, which is set by  $|E|$ . A policymaker can tune  $C$  on accumulation-speed and per-series-cost preferences without affecting the manipulation-gain ceiling. At Capped’s published defaults ( $|E| = 22$ ,  $\kappa = 0.3$  play-in or 0.2 standard), the bound is  $\Delta p_{\text{capped}} \leq 0.3/22 \approx 1.36\%$  play-in and  $0.2/22 \approx 0.91\%$  for any subsequent series.

## 4.5 Three Closures of Assumption (4)

The pool-manipulation gap admits three closure strategies. *Empirical*: Highley et al. [2] simulate 1,000 Bradley-Terry seasons and observe maximum gain  $\sim 3\%$  (statistical, no parameter-explicit ceiling). *Behavioral*: the companion formal-proof video to [2] introduces  $G > \text{expected pool-manipulation gain}$  (where  $G$  is the value of a regular-season win), eliminating the channel via preference structure. *Analytic (this paper)*: Theorem 2 gives a closed-form bound conditioned on structural properties of the index distribution, parameter-explicit and admitting sensitivity analysis (Figure 1a). The three compose; any practical deployment can rely on whichever best matches the available evidence.

# 5 Empirical Validation

We instantiate the configuration of §3 against the ZenGM Basketball-GM engine. §5.1 discloses the simulation model; §5.2 describes the dial grid, objectives, and Pareto methodology; §5.3 reports the headline (Capped@150 as the only Pareto-frontier named variant), the empirical cap-cancels-out confirmation, the reset-on-championship sub-finding, and sensitivity at  $N = 30, 100$ ; §5.4 carries the 26-season historical backtest.

## 5.1 Simulation Model

The ZenGM Basketball-GM engine implements all five COLA variants and the standard NBA lottery; we use it as our reference implementation. ZenGM’s browser-coupled `createLeague()` path does not port to Node, so the testbed bypasses it and synthesizes regular-season outcomes: strength-weighted win records and a single-elimination playoff bracket. The ZenGM lottery code is unchanged: draws use `draft.genOrder` and per-team `cola` updates from `cola.ts`.

Team strength persists across seasons via a Markov model tied to draft outcomes. Let  $s_t \in [0, 1]$  denote strength in season  $t$ ,  $\rho$  the persistence parameter,  $\mu$  the long-run parity mean,  $\alpha$  the per-draft impact,  $\sigma$  the annual shock standard deviation,  $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ , and  $\pi(p) = \max(0, (16 - p)/15)$  the pick-value function (pick 1 contributes full  $\alpha$ , pick 15 zero):

$$s_{t+1} = \text{clip}(\rho s_t + (1 - \rho) \mu + \alpha \pi(p_t) + \varepsilon_t, 0, 1), \quad (2)$$

with  $\rho = 0.9$ ,  $\mu = 0.5$ ,  $\alpha = 0.15$ ,  $\sigma = 0.05$ , and initial strength from `Uniform[0.3, 0.7]`. This preserves the feedback loop the primary objective (worst-franchise gap between conference-finals appearances) is designed to measure. The model is parametrically calibrated for plausibility, not fit to historical team-strength trajectories; formal calibration against league rating systems or Basketball Reference strength-of-schedule data is future work.

## 5.2 Testbed Setup

The headline sweep grid-searches three dials of Definition 2: eligibility  $E$ , cap  $C$ , and carry-over scope  $S$ . The remaining four ( $\Delta, \rho, W, T$ ) are held at canonical settings each variant inherits from Highley et al. [2] and the Substack series. The grid:

- $E \in \{14, 22, 16\text{-tiered}\}$ : 14-team non-playoff pool (Classic, Status quo); 22-team broad pool incl. round-1 losers (Simple, Capped); 16-team tiered pool of the 3-2-1 proposal.
- $C \in \{\text{uncapped}, 100, 150, 200\}$ : uncapped reproduces Classic / Simple;  $C = 150$  is the Capped default [3].
- $S \in \{\text{single-season}, \text{unbounded}, \text{bounded-30yr}, \text{reset-on-championship}\}$ : single-season collapses to 3-2-1 / Status quo; unbounded reproduces Classic; bounded-30yr is the Capped default; reset-on-championship clears every team’s index when each season’s champion is decided.

The grid yields  $3 \times 4 \times 4 = 48$  configurations. The headline run executes each under 50 replicates (seeds 1000–1049, deterministic via overridden `Math.random`) of 30 simulated seasons, for 2,400 (`config`, `replicate`) rows.

Each row reports four objectives. The three dominance dimensions: (i) maximum years between conference-finals appearances for any franchise (lower is better; the primary parity objective, suggested by Prof. Highley); (ii) the analytical manipulation-gain bound  $\Delta p$  as a percentage (lower is better; closed-form from  $E$  and  $C$ , no Monte Carlo); (iii) rank-1-to-5 lottery spread (higher is better; gap between expected picks of the worst and fifth-worst teams). The fourth column is the per-series cost ceiling from Lemma 3 (not used for dominance; N/A for uncapped configurations).

A configuration is Pareto-optimal under 3-objective universal dominance: no other is weakly better on all three dimensions and strictly better on at least one. Per-configuration summaries are medians across the 50 replicates (the median is less sensitive to extreme outliers in max-gap statistics).

### 5.3 Pareto Frontier

With three objectives in play, no single configuration optimizes all of them. The Pareto frontier identifies the configurations a league office could defensibly choose: those not dominated on every objective by any alternative. Configurations off the frontier are strictly dominated and can be discarded; configurations on the frontier represent legitimate trade-offs the design discussion must weigh.

As a concrete dominance check: Status quo (cid 0) has max-years-between-CF of 28.5, manipulation gain of 28.57%, and rank-1-to-5 spread of 2.77. Capped COLA at  $\text{MAX} = 150$  (cid 26) has 28.0, 1.36%, and 2.87 respectively. Capped@150 dominates Status quo on every objective, so Status quo cannot be on the frontier. The same dominance check rules out Classic, Simple, and the 3-2-1 proposal.

Five of the 48 configurations are Pareto-optimal under 3-objective universal dominance (Table 3). Among named variants, only Capped COLA at  $\text{MAX} = 150$  (cid 26) is on the frontier; Status quo (cid 0), Classic (cid 1), Simple (cid 17), and 3-2-1 (cid 32) are all dominated. Capped COLA achieves the joint minimum manipulation gain of 1.364% and the maximum rank-1-to-5 spread of 2.867 on the frontier, dominating Status quo on every objective. Figure 2 visualizes the dominance picture in the (max-yrs-CF, manipulation-gain) plane.

The 2026 cross-variant choice collapses to a single Pareto-relevant named variant under any league weighting over the three dominance objectives. The frontier itself is not a singleton: four unnamed configurations (cids 11, 19, 31, 45) trade max-yrs-CF for manipulation gain or rank-spread, sweeping a strip in the dial space worth further exploration.

**Cap-cancels-out: empirical confirmation.** Corollary 4 establishes that  $C$  drops out of the Capped manipulation-gain bound:  $\Delta p_{\text{capped}} \leq \kappa/|E|$ . The sweep confirms this across the grid’s 36 capped configurations (three pool settings, three cap values, four scopes). All 12 at  $|E| = 22$  realize  $\Delta p = 1.364\%$  (closed form  $0.3/22$ ); all 12 at  $|E| = 14$  realize 2.143%; all 12 at  $|E| = 16$  tiered realize 1.875%. Realized gain is invariant across  $C \in \{100, 150, 200\}$  within each setting; on- and off-frontier capped configurations

| cid                                                                         | $E$       | $C$      | $S$                   | max-yrs-CF | $\Delta p$ [%] | rank-spread | named variant |
|-----------------------------------------------------------------------------|-----------|----------|-----------------------|------------|----------------|-------------|---------------|
| 11                                                                          | 14        | 150      | reset-on-championship | 27.0       | 2.143          | 2.433       |               |
| 19                                                                          | 22        | uncapped | reset-on-championship | 26.0       | 18.182         | 2.650       |               |
| 26                                                                          | 22        | 150      | bounded-30yr          | 28.0       | 1.364          | 2.867       | Capped@150    |
| 31                                                                          | 22        | 200      | reset-on-championship | 27.5       | 1.364          | 2.667       |               |
| 45                                                                          | 16-tiered | 200      | unbounded             | 27.5       | 1.875          | 2.700       |               |
| <b>Named variants under unified units (all dominated except Capped@150)</b> |           |          |                       |            |                |             |               |
| 0                                                                           | 14        | uncapped | single-season         | 28.5       | 28.571         | 2.767       | Status quo    |
| 1                                                                           | 14        | uncapped | unbounded             | 27.5       | 28.571         | 2.367       | Classic       |
| 17                                                                          | 22        | uncapped | unbounded             | 29.0       | 18.182         | 2.667       | Simple        |
| 32                                                                          | 16-tiered | uncapped | single-season         | 28.0       | 25.000         | 2.567       | 3-2-1         |

Table 3: The five Pareto-optimal configurations (top block; cids 11, 19, 26, 31, 45) and the five named variants under unified units (bottom block; rows show medians across 50 replicates of 30 simulated seasons each, seeds 1000–1049). Capped COLA at  $\text{MAX} = 150$  (cid 26) is the only named variant on the Pareto frontier; the other four named variants are universally dominated. Capped@150 achieves the minimum manipulation gain (tied with cid 31 at 1.364%) and the maximum rank-1-to-5 spread (2.867) on the frontier.

share the same per- $|E|$   $\Delta p$ . A league can choose  $C$  on per-team accumulation speed and per-series tanking cost (per Lemma 3, the per-series cost is  $0.3C$ ) without affecting the manipulation ceiling, fixed by  $|E|$ . To lower the ceiling, move  $E$ , not  $C$ .

**Reset-on-championship as a winning scope.** Three of the five Pareto-optimal configurations (cids 11, 19, 31) use the reset-on-championship scope; one uses bounded-30yr (cid 26, Capped@150) and one unbounded (cid 45). The single-season scope does not appear on the frontier; the 3-2-1 proposal (cid 32) is dominated. Reset-on-championship is event-based: the league champion’s identity is observable at one point each season and the reset triggers across the entire pool at that point. The scope retains multi-year accumulation while preventing the index inflation single-team unbounded accumulation would produce. Bounded-30yr (cid 26) is the cap-driven analog; unbounded (cid 45) appears on the frontier only when paired with the 16-tiered pool, which structurally limits per-team accumulation via the tiering. §6.3 interprets the event-based scope as a policy lever.

**Sensitivity at  $N = 30$  and  $N = 100$ .** A subset combining the Pareto-optimal set with the five named variants (9 deduplicated configurations) evaluated at  $N = 30$  and  $N = 100$  shifts the primary-objective mean by less than 10% from  $N = 30$  to  $N = 100$ . The headline  $N = 50$  is statistically sufficient for the dominance ordering. Within-subset Pareto recomputation at the two settings produces small membership flips ( $N = 30$ :  $\{0, 11, 19, 26, 45\}$ ;  $N = 100$ :  $\{31\}$ ) that reflect within-subset re-ordering rather than full-grid instability; the headline  $N = 50$  frontier  $\{11, 19, 26, 31, 45\}$  is canonical. The closed-form manipulation-gain dimension is invariant across  $N$  by construction.

## 5.4 Historical Backtest

The backtest computes COLA state for every team across 26 NBA seasons (1999–2025) using verified Basketball Reference data: 775 team-season records, 776 first-round draft picks, and 55 lottery-day-attributed pre-draft pick trades. The engine separates pre-draft state (Phase A, used for draft-probability calculations) from post-draft diminishment (Phase B, carried forward to the next season). Cold-start effects stabilize by approximately season 5; we report aggregates over 21 stable seasons (2005–2025). The backtest is *static*: it applies COLA rules to actual historical outcomes. Under a real COLA regime, draft outcomes would differ

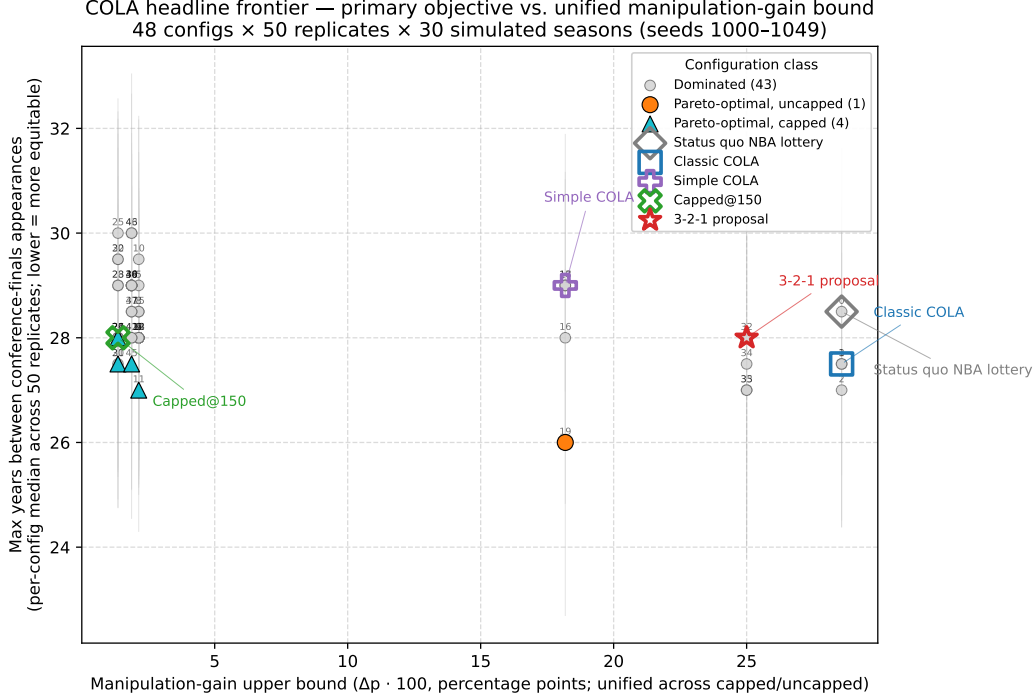


Figure 2: Headline Pareto frontier in the (max-yrs-CF, manipulation-gain) plane. Y-axis: median maximum years between conference-finals appearances across 50 replicates (lower is better). X-axis: closed-form manipulation gain  $\Delta p$  as a percentage (lower is better). The X-axis spans two analytical regimes: uncapped configurations sit at  $\Delta p \approx 4/|E|$  (Classic / Simple / Status quo / 3-2-1); capped configurations collapse to  $\Delta p \approx 0.3/|E|$  (Corollary 4), creating the tight band at the left. The third dominance dimension (rank-1-to-5 spread) is not visualized; see Table 3 for the full universal-dominance set.

in ways that change future drought lengths and indices. We caveat the comparative cross-variant claims accordingly. An audit of all 364 lottery picks (14 picks  $\times$  26 seasons) identified 8 instances where the draft-day holder differed from the lottery-day holder via post-lottery trades; COLA uses *lottery-day* attribution.

The 2013–14 through 2016–17 Philadelphia 76ers are the canonical strategic-tanking sequence in modern NBA history: four consecutive top-3 lottery outcomes (picks #3, #3, #1, #3) acquired through deliberate multi-season non-competitiveness. Under the current weighted lottery, each tanked season yielded a top-tier pick. Under Classic COLA, graduated diminishment makes the same sequence self-defeating. Philadelphia’s Classic COLA trajectory: 2,750 in 2013–14 (rank 8/14)  $\rightarrow$  2,375 in 2014–15 (9/14)  $\rightarrow$  2,188 in 2015–16 (10/14)  $\rightarrow$  1,000 in 2016–17 (14/14). Each top-3 pick triggers graduated diminishment (#1: full reset; #3: 50% reduction), leaving Philadelphia at the bottom of the lottery pool by the fourth tanking season despite a 10–72 record. The Sacramento Kings hold the top Classic COLA index throughout this period, climbing from 7,250 in 2013–14 to 10,250 in 2016–17 via organic non-competitiveness. The Classic diminishment schedule ( $\{1.0, 0.75, 0.5, 0.25\}$  for picks #1–4) dominates the  $\alpha = 1,000$  annual accumulation; a team that receives a top-3 pick in year  $t$  cannot re-enter the top of the lottery in year  $t + 1$  through tanking alone.

The Capped cap calibration sweep over  $\text{MAX} \in \{75, 100, 125, 150, 175, 200\}$  across the 21 stable seasons replicates the published default of  $\text{MAX} = 150$  [3]: at  $\text{MAX} = 150$ , a mean of 1.62 teams sit at the cap (range 0–3), matching the design target of fewer than two teams at the maximum on average. Sub-100 caps over-compress the rank-1-vs-rank-5 separation; caps above 175 leave fewer than one team at the cap on average. The static backtest is a second-tier validation alongside the headline Pareto result: it confirms the qualitative comparative-statics the simulated sweep reports under closed-loop dynamics. Trade han-

ding is a practical implementation question for any deployment; following Highley [11], we recommend original-holder diminishment paired with a strengthened Stepien Rule.

## 6 Discussion and Open Directions

Theorem 2 bounds Classic COLA’s open assumption (4) at approximately 4% under empirically calibrated parameters, with explicit sensitivity to  $T_{\max}$ ,  $T_{\text{boundary}}$ ,  $n$ ,  $\bar{k}$ . Lemma 3 bounds Capped’s per-series exposure. Combined, the four manipulation channels under Capped (single-season tanking, pool manipulation, multi-year strategic tanking, per-series playoff tanking) compose to a small joint exposure relative to the status-quo lottery, where repeat tanking has memoryless returns.

### 6.1 3-2-1 as 2026 Stopgap, COLA as 2029 Candidate

We read 3-2-1 as a short-horizon stopgap: it accepts the Munro-Banchio tradeoff by flattening odds, with the bottom three teams given fewer balls than the middle of the pool. The proposal’s three-year re-evaluation provision reopens the design discussion in 2029. COLA is the natural 2029 candidate on a different axis: dial  $S$  (carry-over scope) shifts the priority basis to multi-year playoff history, an option neither 3-2-1 nor any single-season mechanism considers. 3-2-1 sacrifices differential reward for non-tanking; COLA preserves differential reward by relocating it to a non-manipulable basis. The 2029 discussion benefits from having both options analytically characterized.

### 6.2 Dial-Space Adoption Tradeoffs

Each dial matters more to some audiences than others. *Fans and media* care most about  $E$  and  $W$ : the current 14-team weighted lottery fits a 30-second broadcast graphic, while Capped’s exclusion-rule pool and Countdown’s nested Monte Carlo do not. *General managers and owners* care most about  $\Delta$  and  $\rho$ : a multi-year rebuild needs a forecastable trajectory. Classic’s flat  $\Delta = \alpha$  and graduated  $\rho$  are the most tractable; Capped’s wins-based  $\Delta$  and six-step  $\rho$  add complexity but remain forecastable; Countdown’s  $\Delta = d \cdot w$  multiplies factors that move in opposite directions for a rebuilding team and produces noisier multi-year forecasts. *League counsel* cares most about  $S$ : multi-year scopes raise questions about mid-season team relocation, ownership change, and league realignment that single-season scopes avoid. The NBA’s apparent preference for single-season scopes in 3-2-1 is consistent with  $S$  being the dial under the most external pressure.

### 6.3 Reset-on-Championship as a Policy Lever

The sweep identifies a third option for  $S$  beyond time-bounded windows and unbounded accumulation: an outcome-triggered reset. For league counsel, reset-on-championship bounds the carry-over horizon without imposing a calendar deadline, and the trigger event is already part of the league’s annual cadence. For GMs, it preserves the multi-year rebuild trajectory until the rebuild succeeds; winning the championship resets the priority index. Event-based scopes extend the design space beyond time-based alternatives, and the sweep validates them as competitive.

**Open directions:** three follow-on questions extend this work. (i) *Tighter manipulation bound*: the structural anti-correlation in Definition 3 is a calibrated assumption rather than a derived consequence of the mechanism. A stationarity argument bounding  $T_{\text{boundary}}$  as a function of league parameters (team count, season length, playoff structure) would close assumption (4) entirely. (ii) *Pick-value gap*: combining the

probability bound with a bound on the pick-value gap  $d_k - d_{k+1}$ , derived from the distribution of draft-eligible prospect quality, would close the manipulation incentive in absolute terms. (iii) *Axiomatic embedding*: the lift from Coreno and Balbuzanov [4]’s axioms (RP, EF1, SP) to multi-period priority mechanisms remains open; a formal lift would embed COLA in the assignment-mechanism literature and clarify which one-period impossibilities transfer.

A reproducibility artifact accompanies this paper: an interactive backtester implementing all five variants plus the 3-2-1 proposal across 26 NBA seasons, with locked-seed Monte Carlo for Countdown probabilities [12].

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